Arithmetic constructions of hyperbolic Kac-Moody groups

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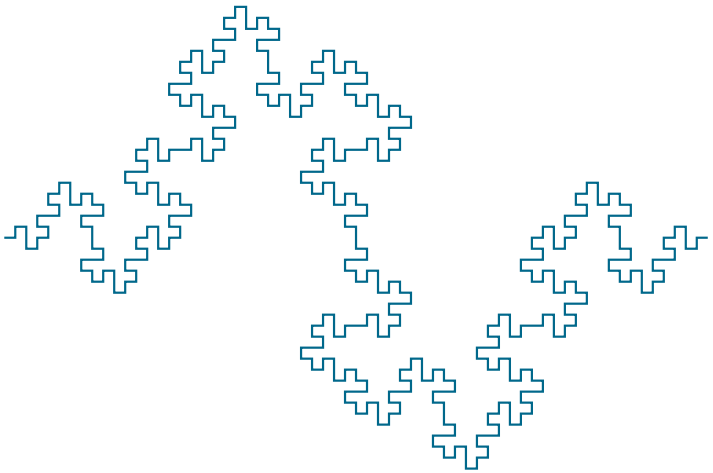

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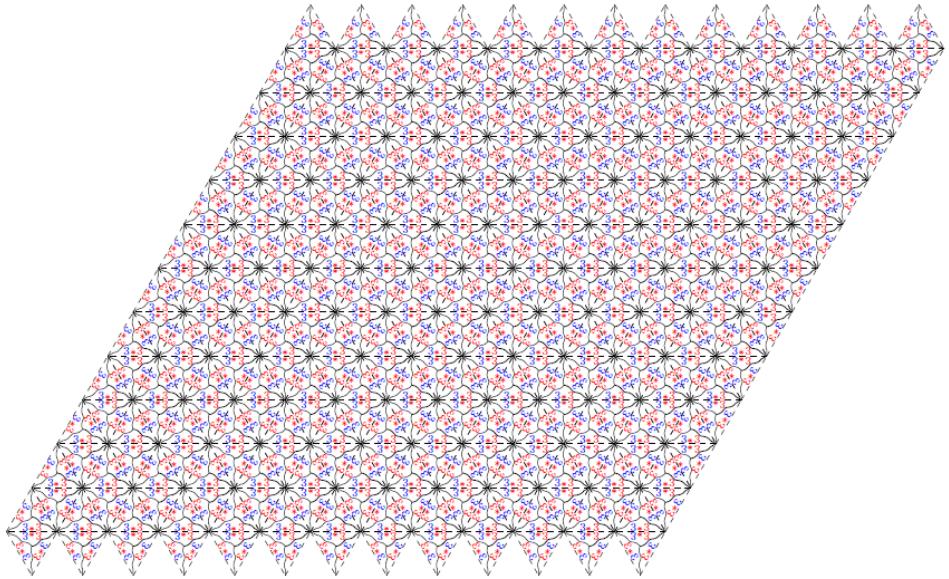
Objective

Kac–Moody groups and algebras are the most natural generalizations to infinite dimensions of finite dimensional simple Lie groups and Lie algebras.

In finite dimensions, arithmetic subgroups are discrete groups of matrices with integer entries.

Here we consider the infinite dimensional analog, by studying Kac–Moody groups over $\mathbb{Z}$.



Overview

We consider and compare two distinct constructions of a simply laced and hyperbolic Kac–Moody group $G = G(R)$ over a commutative ring R with 1.

One of these groups is Tits's construction of $G(R)$, though we work only with a finite presentation of $G(R)$ obtained by Allcock and C.

The other is a Kac–Moody group $G^{\lambda}(R)$ constructed by C, C–Garland, using an integrable highest weight module V^{λ} (with dominant integral weight λ) for the underlying Kac–Moody algebra. This is the infinite dimensional analog of an *elementary Chevalley group* over the ring R, namely, a group generated by real root group generators.

The affine case was studied in detail in the foundational work of Howard Garland.

Kac–Moody algebras: Introductory remarks

Let $\mathfrak{g}$ be a Kac–Moody algebra with Cartan subalgebra $\mathfrak{h}$ and root space decomposition:

$$\mathfrak{g} = \mathfrak{g}^+ \oplus \mathfrak{h} \oplus \mathfrak{g}^-$$

The roots $\Delta \subset \mathfrak{h}^*$ are the eigenvalues, and $\mathfrak{g}^\pm = \oplus_{\alpha \in \Delta^\pm} \mathfrak{g}_\alpha$ where the root spaces

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} \mid [h, x] = \alpha(h)x,\ h \in \mathfrak{h}\}$$

are the corresponding eigenspaces. When $\mathfrak{g}$ is infinite dimensional, $|\Delta| = \infty$.

In this case, $\mathfrak{g}$ has 2 types of roots: *real roots* with positive norm and *imaginary roots* with negative or zero norm.

Some elementary examples

Here are some symmetric Cartan matrices which encode data for the root system and the generators and relations of various Lie algebras.

$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, the finite dimensional Lie algebra of type A_2

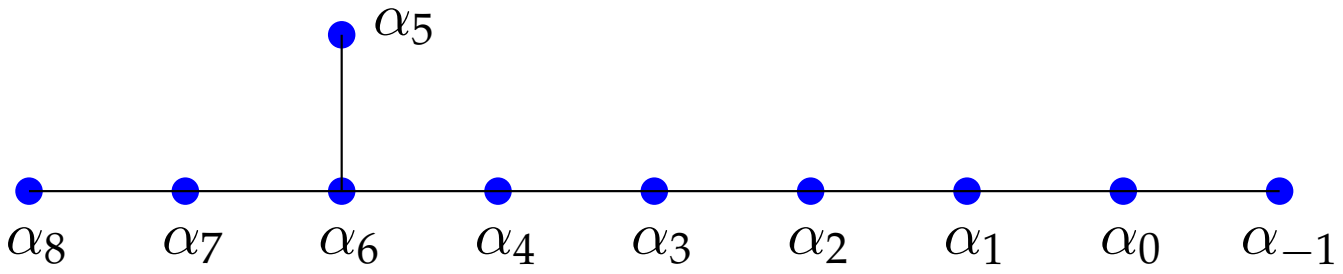
$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, the rank 2 affine Kac–Moody algebra $A_1^{(1)}$

$\begin{pmatrix} 2 & -m \\ -m & 2 \end{pmatrix}_{m \in \mathbb{Z}_{\geq 3}}$, the rank 2 hyperbolic Kac–Moody algebra $H(m)$

$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$, the rank 3 hyperbolic Feingold–Frenkel algebra

Main example: The hyperbolic Kac–Moody algebra $\mathfrak{e}_{10}$

An extension of the Dynkin diagram of the finite dimensional Lie algebra $\mathfrak{e}_8$ gives the affine diagram $\mathfrak{e}_9$, which was discovered by Tits in the early 1960's.



A further extension of the Dynkin diagram of $\mathfrak{e}_9$ gives the hyperbolic Dynkin diagram $\mathfrak{e}_{10}$.

There are fundamental open questions about the mathematical structure of $\mathfrak{e}_{10}$, as well as much interest and intrigue due to the apparent appearance of $\mathfrak{e}_{10}$ symmetry in supergravity and superstring theories.

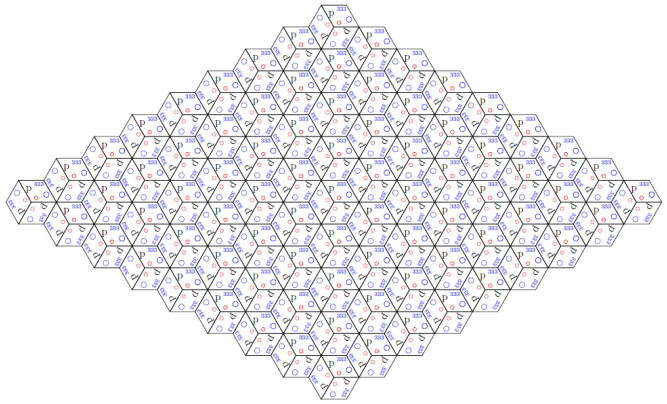
Kac–Moody groups: First remarks

A Kac–Moody group is an abstract group G associated to a Kac–Moody algebra $\mathfrak{g}$.

We view the group G as an infinite dimensional Lie group, though there is no well–defined notion of dimension in the currently known Kac–Moody group constructions.

Over general (commutative) rings, the final definition of a Kac–Moody group has not yet been found—it should be some sort of generalization of the Chevalley–Demazure group scheme.

Some groundwork for establishing the appropriate notion has been given by Tits, Allcock, C.

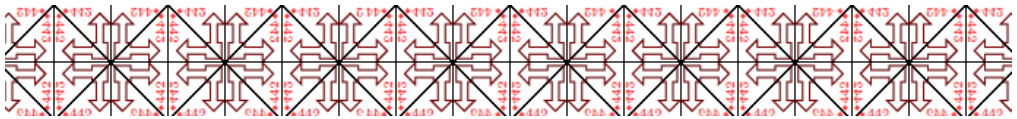


Kac–Moody groups over fields

The methods for constructing Kac–Moody groups over fields, however, are numerous (axiomatic, algebro–geometric, representation–theoretic, generators and relations, Hilbert manifolds and line bundles, Banach completions, Chevalley type constructions,...)

Several authors have proposed group constructions using a range of techniques and additional external data (Moody and Teo, Slodowy, Tits, Mathieu, Marcuson, Garland, Kac and Peterson, Pressley and Segal, Goodman and Wallach, Kumar, Rémy and Ronan, Rousseau, C and Garland...)

Many constructions use some version of the *Tits functor*.



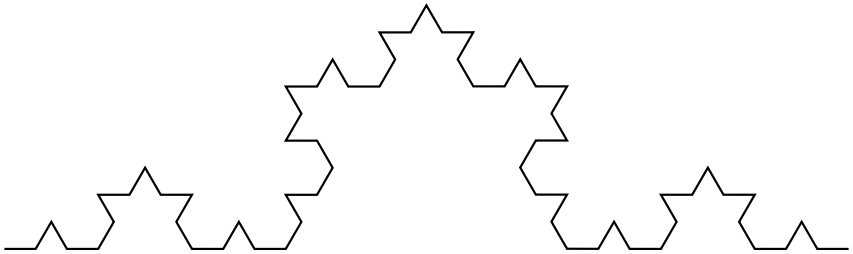
Tits' Kac–Moody group functor

Tits defined a functor $\widetilde{\mathfrak{G}}$ from commutative rings to groups.

He showed that his model of a Kac–Moody group is the natural one, at least for fields.

Namely, any group functor $\mathfrak{G}$ with some obviously desirable properties, stated as axioms KMG1–KMG5, admits a functorial homomorphism $\widetilde{\mathfrak{G}} \to \mathfrak{G}$, which is an isomorphism at every field.

However, Tits did not show that his group $\widetilde{\mathfrak{G}}$ satisfies the axioms KMG1–KMG5.



Tits' Kac–Moody group presentation

Tits also gave generators and relations for his Kac–Moody group $\widetilde{\mathfrak{G}}$, generalizing the Steinberg presentation for finite dimensional Lie groups.

In the finite dimensional case, there is a Chevalley type commutation relation of the form

$$[\chi_\alpha(u), \chi_\beta(v)] = \prod_{m,n} \chi_{m\alpha+n\beta}(C_{mn\alpha\beta} u^m v^n)$$

between every pair of root groups U_α, U_β. Here $u, v \in R$, $C_{mn\alpha\beta}$ are integers (structure constants) and the χ_α are viewed as formal symbols in

$$U_\alpha = \{\chi_\alpha(u) \mid \alpha \in \Delta,\ u \in R\} \cong (R, +).$$

However, in the infinite dimensional case, Tits' presentation of Kac–Moody groups has infinitely many Chevalley commutation relations.

Determining Tits' Kac–Moody group presentation

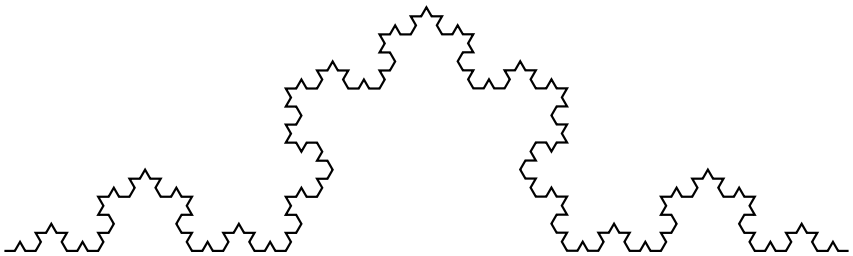
In the infinite dimensional Kac–Moody case, Tits determined that whenever a pair of real roots is 'prenilpotent', then there is a Chevalley commutation relation necessary for defining the Kac–Moody group. In order to make Tits' presentation complete, over a commutative ring R with 1, we need to:

Explicitly describe the prenilpotent pairs of roots.

This usually requires us to:

Explicitly describe the set of positive real roots.

There is no guarantee that either of these tasks can be carried out in practice.



Prenilpotent pairs

Let (α, β) be a pair of real roots and let W denote the Weyl group. Then (α, β) is called a *prenilpotent pair,* if there exist $w,\ w' \in W$ such that

$$w\alpha,\ w\beta \in \Delta_+^{re} \text{ and } w'\alpha,\ w'\beta \in \Delta_-^{re}.$$

A pair of roots $\{\alpha, \beta\}$ is prenilpotent if and only if $\alpha \neq -\beta$ and

$$(\mathbb{Z}_{>0}\alpha + \mathbb{Z}_{>0}\beta) \cap \Delta_+^{re}$$

is a finite set. For every prenilpotent pair of roots $\{\alpha, \beta\}$, Tits defined the Chevalley commutation relation

$$[\chi_\alpha(u), \chi_\beta(v)] = \prod_{m\alpha+n\beta \in (\mathbb{Z}_{>0}\alpha+\mathbb{Z}_{>0}\beta)\cap\Delta_+^{re}} \chi_{m\alpha+n\beta}(C_{mn\alpha\beta}u^m v^n)$$

where $u, v \in R$ and $C_{mn\alpha\beta}$ are integers (structure constants).

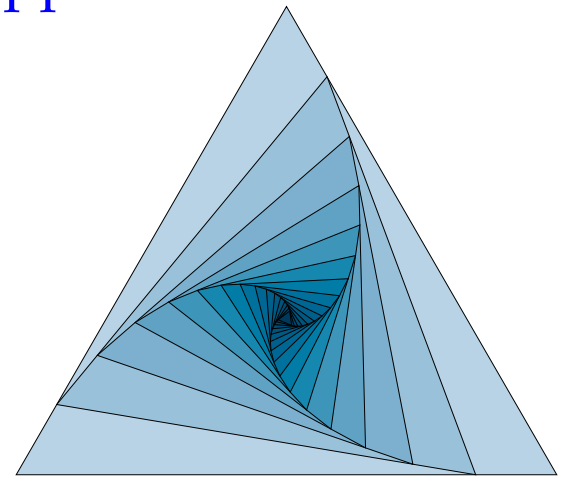
The difficulty of finding prenilpotent pairs for E_{10}

For E_{10}, recent work of Allcock has shown that the problem of determining all prenilpotent pairs is tractable but impractical.

Namely, Allcock showed that for the number of W–orbits of prenilpotent pairs of real roots having inner product equal to k grows at least as fast as $(const)k^7$ as $k \to \infty$.

For each such orbit, we then have to enumerate the prenilpotent pairs of real roots.

We need a different approach.



Tits' presentation for a Kac–Moody group

Tits defined the *Steinberg group*

$$\mathfrak{St}(R) = *_{\alpha \in \Delta^{re}} U_\alpha / \text{Chev. comm. relns. on prenilpotent pairs}$$

Tits' Kac–Moody group $G(R)$ is a quotient of the Steinberg group $\mathfrak{St}(R)$ by the additional relations which define the torus.

Tits' Kac–Moody group has an infinite set of generators and an infinite set of defining relations. The sets of generators and relations are redundant and can be reduced significantly.

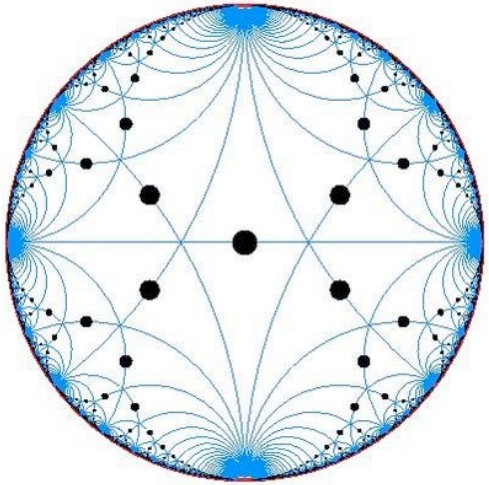
In earlier work, Allcock defined a new functor, the *pre–Steinberg group* $\mathfrak{PSt}(R)$ and showed that in many cases $\mathfrak{PSt}(R) \cong \mathfrak{St}(R)$. The presentation of $\mathfrak{PSt}(R)$ is defined in terms of the Dynkin diagram rather than the full infinite root system.

Joint work with Daniel Allcock

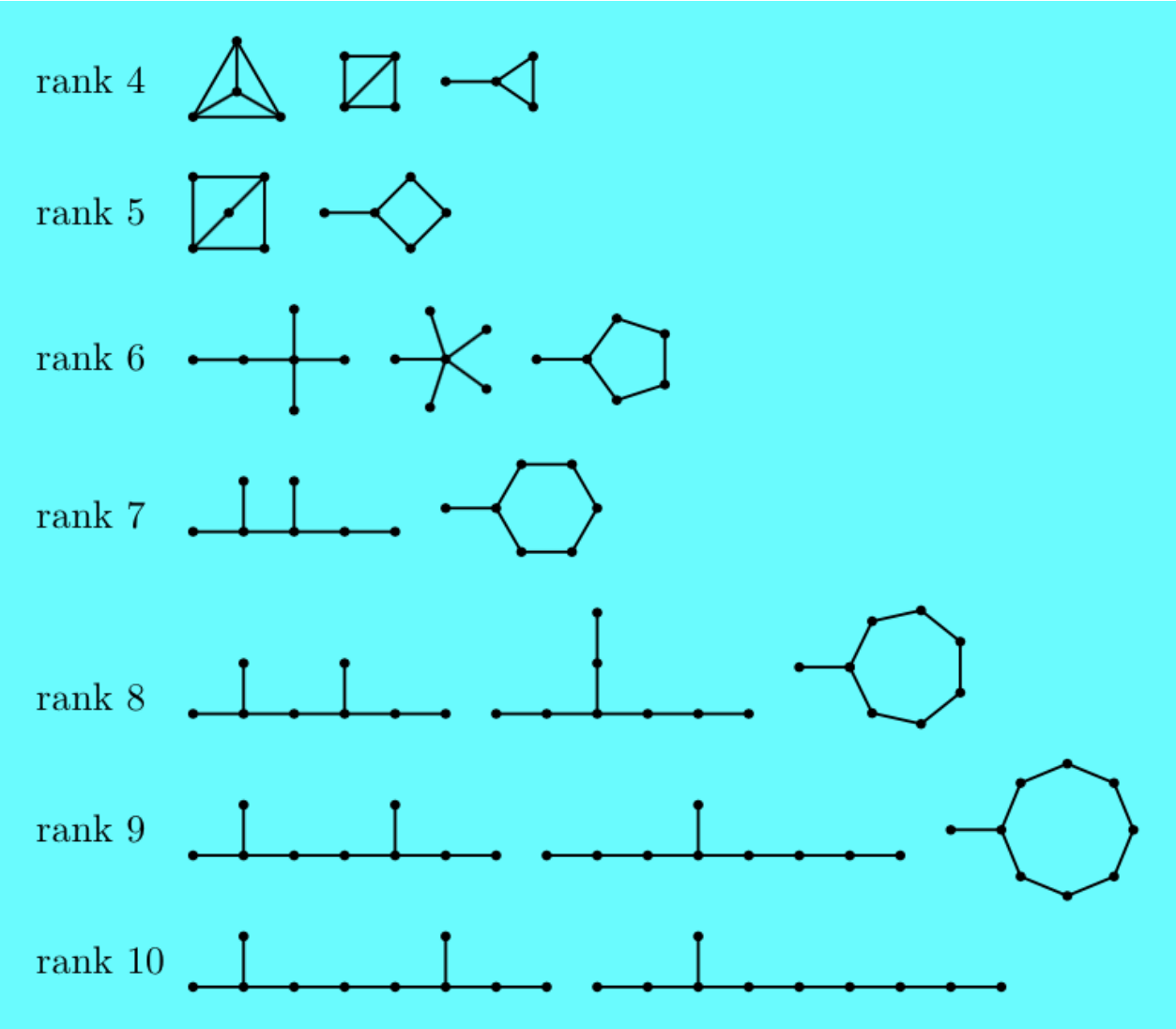
We obtained a further simplification of the isomorphism $\mathfrak{PSt}(R) \cong \mathfrak{St}(R)$ for root systems that are simply laced and hyperbolic.

A Dynkin diagram is *simply laced* if it consists only of single bonds between nodes.

It is *hyperbolic* if it is neither of affine nor or finite dimensional type, but every proper connected subdiadram is either of affine or finite dimensional type.



The simply laced hyperbolic Dynkin diagrams



The isomorphism $\mathfrak{PSt}(R) \cong \mathfrak{St}(R)$

In Allcock's group $\mathfrak{PSt}(R)$, all Chevalley commutation relations are implied by those arising from pairs of simple roots $\{\alpha_i, \alpha_j\}$ on the Dynkin diagram.

The isomorphism $\mathfrak{PSt}(R) \cong \mathfrak{St}(R)$ means that in the Steinberg group $\mathfrak{St}(R)$ (and hence in the Kac–Moody group), the Chevalley commutation relations from the Dynkin diagram imply the Chevalley commutation relations in the whole group.

Our simplification of the isomorphism $\mathfrak{PSt}(R) \cong \mathfrak{St}(R)$ for root systems that are simply laced and hyperbolic uses properties of the hyperbolic Weyl group (as a Coxeter group) and the geometry of its action on (a projectivization of) the root space.

Finitely many defining relations parametrized over R

The Kac–Moody group $G(R)$ is generated by elements $X_i(t)$, $t \in R$, S_i, $i \in \{1, \ldots, \ell\}$. The elements $\widetilde{h}_i(a)$ in the first column of the Table are

$$\widetilde{h}_i(a) := \widetilde{s}_i(a)\widetilde{s}_i(-1),$$

$$\widetilde{s}_i(a) := X_i(a)S_iX_i(a^{-1})S_i^{-1}X_i(a).$$

Any i, all $a, b \in R^\times$	Each $i, t, u \in R$	$i \neq j$ not adjacent $t, u \in R$	$i \neq j$ adjacent $t, u \in R$
$\widetilde{h}_i(a)\widetilde{h}_i(b) = \widetilde{h}_i(ab)$	$X_i(t)X_i(u) = X_i(t+u)$	$S_iS_j = S_jS_i$	$S_iS_jS_i = S_jS_iS_j$ $S_i^2S_jS_i^{-2} = S_j^{-1}$
	$[S_i^2, X_i(t)] = 1$	$[S_i, X_j(t)] = 1$	$X_i(t)S_jS_i = S_jS_iX_j(t)$ $S_i^2X_j(t)S_i^{-2} = X_j(t)^{-1}$
	$S_i = X_i(1)S_iX_i(1)S_i^{-1}X_i(1)$	$[X_i(t), X_j(u)] = 1$	$[X_i(t), X_j(u)] = S_iX_j(tu)S_i^{-1}$ $[X_i(t), S_iX_j(u)S_i^{-1}] = 1$

Table: The defining relations for $G(R)$, G simply laced and hyperbolic, R a commutative ring with 1

A finite presentation for $G(\mathbb{Z})$

Over $R = \mathbb{Z}$, the generators $X_i(u)$ are obtained from $X_i = X_i(1)$ via $X_i(u) = X_i^u$.

Thus we can rewrite the presentation without the scalars from the underlying ring and obtain a *finite* presentation for $G(\mathbb{Z})$:

Any i	Each $i = \{1, \ldots, \ell\}$	$i \neq j$ not adjacent	$i \neq j$ adjacent
$S_i^4 = 1$	$[S_i^2, X_i] = 1$	$S_iS_j = S_jS_i$	$S_iS_jS_i = S_jS_iS_j$ $S_i^2S_jS_i^{-2} = S_j^{-1}$
	$S_i = X_iS_iX_iS_i^{-1}X_i$	$[S_i, X_j] = 1$	$X_iS_jS_i = S_jS_iX_j$ $S_i^2X_jS_i^{-2} = X_j^{-1}$
		$[X_i, X_j] = 1$	$[X_i, X_j] = S_iX_jS_i^{-1}$ $[X_i, S_iX_jS_i^{-1}] = 1$

Table: The defining relations for $G(\mathbb{Z})$, G simply laced and hyperbolic

The Feingold–Frenkel hyperbolic Kac–Moody algebra AE_3

Consider the hyperbolic Kac–Moody algebra with generalized Cartan matrix

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}$$

The Weyl group W is the $(3, 2, \infty)$-triangle group:

$$W = \langle w_1, w_2, w_3 \mid w_i^2 = 1,\ (w_2 w_3)^3 = 1, (w_1 w_3)^2 = 1 \rangle,$$

which is isomorphic to $PGL_2(\mathbb{Z})$.The Dynkin diagram

is not simply laced and hence does not satisfy the conditions to give rise to our finite presentation. In recent work with Scott Murray, we give a recursive method for determining the infinite set of prenilpotent pairs of real roots in AE_3.

Representation–theoretic Kac–Moody group

Theorem ([C]) Let $\mathfrak{g}$ be a symmetrizable Kac–Moody algebra over a commutative ring R with 1. Let α_i be the simple roots and e_i, f_i the Chevalley generators of $\mathfrak{g}$, for $i = 1, \ldots, \ell$. Let V_R^λ be an R–form of an integrable highest weight module V^λ for $\mathfrak{g}$, with highest weight λ and defining representation $\rho : \mathfrak{g} \to End(V_R^\lambda)$. Let

$$G^\lambda(R) = \langle exp(\rho(se_i)),\ exp(\rho(tf_i)) \mid s, t \in R \rangle$$

be the corresponding representation–theoretic Kac–Moody group. Let $s, t \in R$, $u \in R^\times$ and set

$$\chi_{\alpha_i}(s) \ = \ exp(\rho(se_i)), \quad \chi_{-\alpha_i}(t) \ = \ exp(\rho(tf_i)),$$

$$\widetilde{w}_{\alpha_i}(u) = \chi_{\alpha_i}(u)\chi_{-\alpha_i}(-u^{-1})\chi_{\alpha_i}(u), \quad h_{\alpha_i}(u) \ = \ \widetilde{w}_{\alpha_i}(u)\widetilde{w}_{\alpha_i}(1)^{-1}.$$

Then $G^\lambda(R)$ has the following generating sets:

(1) $\chi_{\alpha_i}(s)$ and $\chi_{-\alpha_i}(t)$,

and

(2) $\chi_{\alpha_i}(s)$ and $\widetilde{w}_{\alpha_i}(1) = \chi_{\alpha_i}(1)\chi_{-\alpha_i}(-1)\chi_{\alpha_i}(1)$.

A similar construction for G^λ over arbitrary fields was used in earlier work with Garland to construct Kac–Moody groups over finite fields.

Arithmetic subgroup

When $R = \mathbb{C}$ we define the integral subgroup $G^\lambda(\mathbb{Z})$ to be the group

$$G^\lambda(\mathbb{Z}) = \langle \chi_{\alpha_i}(s), \chi_{-\alpha_i}(t) \mid s, t \in \mathbb{Z},\ i \in I \rangle.$$

It is non–trivial to prove that this group coincides with the 'Chevalley group' over $\mathbb{Z}$, namely the subgroup of $G^\lambda(\mathbb{C})$ preserving the lattice $V^\lambda_{\mathbb{Z}}$ in the representation space V^λ.

Subtle issues: If $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ is written in terms of the generators

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \chi_{\pm\alpha}(t_1)\chi_{\pm\alpha}(t_2)\dots\chi_{\pm\alpha}(t_k)$$

then it is not necessarily the case that the scalars t_i are all integers:

$$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \chi_\alpha(\frac{1}{2})h_\alpha(\frac{1}{2})\chi_{-\alpha}(\frac{1}{2}) = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix}.$$

Uniqueness of complete Kac–Moody groups over fields

Let R be a field. In earlier work, Garland constructed affine Kac–Moody groups as central extensions of loop groups and characterized the dependence of a completion of $G^\lambda(R)$ on a choice of dominant integral weight λ in terms of the Steinberg cocycle.

Let $G(R)$ denote Tits' Kac–Moody group and $G^\lambda(R)$ denote the representation–theoretic Kac–Moody group constructed by C–Garland.

Let $K^\lambda(R) = Ker(\rho_\lambda : G(R) \to G^\lambda(R))$. There are a number of known results which determine the natural extension of K^λ to completions of $G(R)$ and $G^\lambda(R)$ when R is a field.

Uniqueness of complete Kac–Moody groups over fields

For symmetrizable Kac–Moody groups over fields R, Baumgartner and Rémy showed that K^λ is contained in the center of a completion of $G(R)$.

Recent work of Rousseau shows that over fields of characteristic zero, completions of $G(R)$ and $G^\lambda(R)$ are isomorphic as topological groups.

The completion of $G^\lambda(R)$ by C–Garland is constructed with respect to a choice of dominant integral weight λ. Thus Rousseau's work also shows that over fields of characteristic zero, the complete Kac–Moody groups of C–Garland are independent of the choice of λ up to isomorphism, as topological groups.

Uniqueness of Kac–Moody groups over $\mathbb{Z}$?

Let G be simply laced and hyperbolic. We consider a map from the finite presentation of $G(R)$ of Allcock–C to the rep. theoretic group $G^\lambda(R)$, over a commutative ring R with 1.
We define the map ρ_λ:

$$G(R) \xrightarrow{\rho_\lambda} G^\lambda(R)$$

from generators of $G(R)$ to generators of $G^\lambda(R)$:

$$X_i(t) \mapsto \chi_{\alpha_i}(t)$$

$$S_i \mapsto \widetilde{w}_{\alpha_i}$$

$$\widetilde{h}_i(u) \mapsto h_{\alpha_i}(u)$$

for $t \in R$, $u \in R^\times$, $i \in \{1, \ldots, \ell\}$.

Theorem. (C–Wagner) *When $R = \mathbb{Z}$, this map extends to a group homomorphism $G(\mathbb{Z}) \to G^\lambda(\mathbb{Z})$* (*up to a sign in the structure constants).

The kernel of $\rho_{\lambda,\mathbb{Z}}$

Over $R = \mathbb{Z}$, we study the *minimal* or incomplete Kac–Moody group, where the methods for determining K^λ are less transparent.

Theorem (C–Wagner) Let G be simply laced and hyperbolic. Then the kernel K^λ of the map $\rho_{\lambda,\mathbb{Z}} : G(\mathbb{Z}) \to G^\lambda(\mathbb{Z})$ lies in

$$Center(G(\mathbb{C})) \leq H(\mathbb{C}) = \langle h_{\alpha_i}(u) \mid u \in \mathbb{C}^\times \rangle.$$

If further the natural group homomorphism $\varphi : G(\mathbb{Z}) \to G(\mathbb{C})$ is injective, then $K^\lambda \leq H(\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^{rank(G)}$, a *finite* group.

Injectivity of the natural map $\varphi : G(\mathbb{Z}) \to G(\mathbb{C})$ is not currently known and depends on functorial properties of Tits' group.

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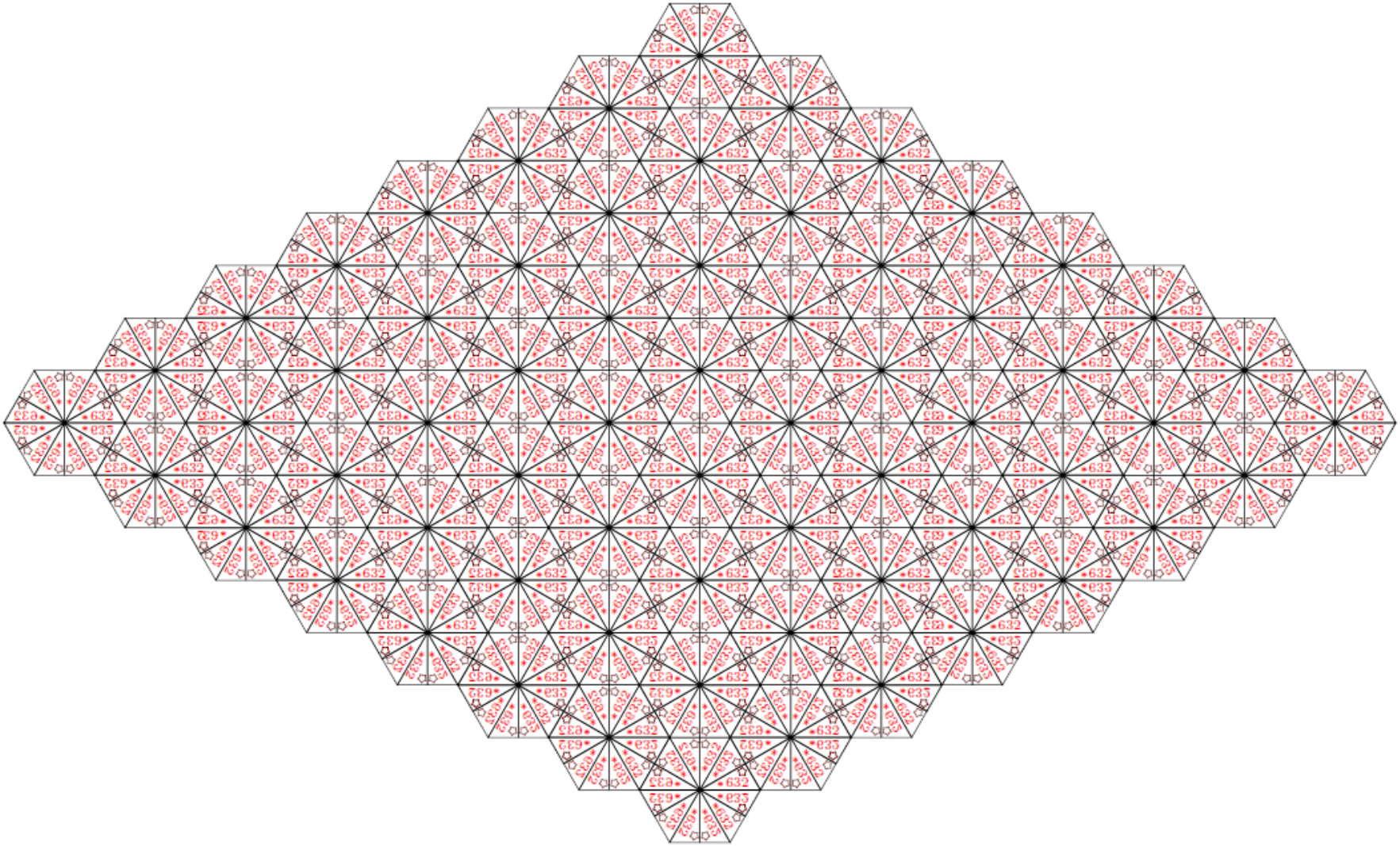
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